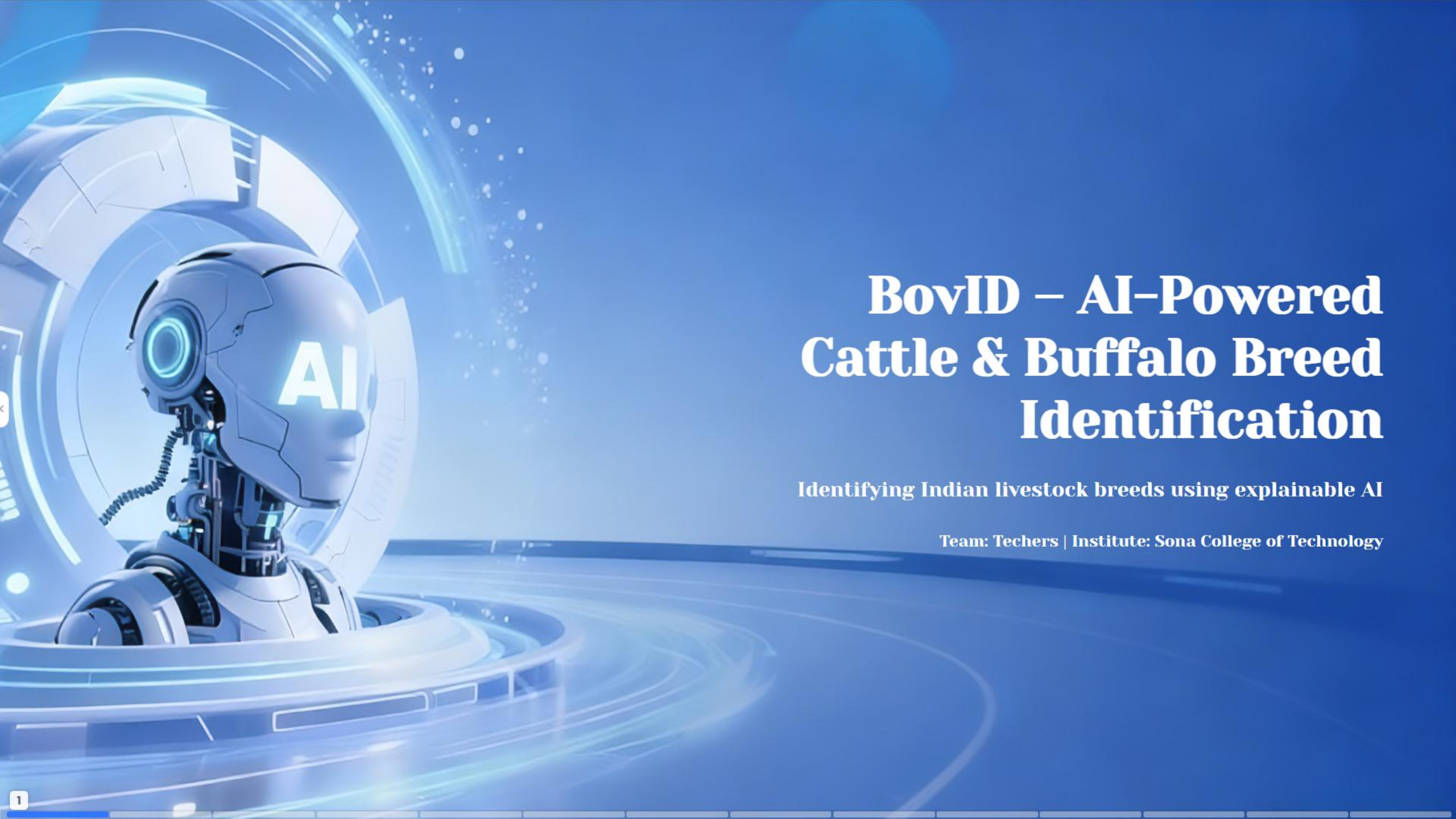




BovID – AI-Powered Cattle & Buffalo Breed Identification

Identifying Indian livestock breeds using explainable AI

Team: Techers | Institute: Sona College of Technology

Problem Statement

Manual, Expert-Dependent, Error-Prone Identification

India has 74+ registered cattle & buffalo breeds, yet farmers still rely on manual methods leading to misidentification, lost subsidies, and poor breeding decisions.



Farmers Misidentify Breeds

Lose access to correct subsidies and make poor breeding choices.



No Digital Solution

No scalable, farmer-friendly tool exists for accurate breed identification.



Rural Accessibility Gap

Experts are scarce in remote villages where identification is most needed.

Why This Problem Matters

30%

Rural Income Loss

Livestock Livelihood Impact

- Livestock contributes significantly to rural household income, making accurate breed identification crucial for economic stability.
- Incorrect ID leads to lower milk productivity
 - Threatens indigenous breed conservation
 - Causes inefficient subsidy distribution

25%

Milk Productivity Drop

Expert Accessibility Gap

- Veterinary experts are concentrated in urban areas, leaving rural farmers without reliable identification support.
- Technology can bridge this urban-rural divide
 - Mobile solutions reach remote villages
 - AI democratizes expert knowledge

40%

Genetic Dilution Risk

Policy Implementation

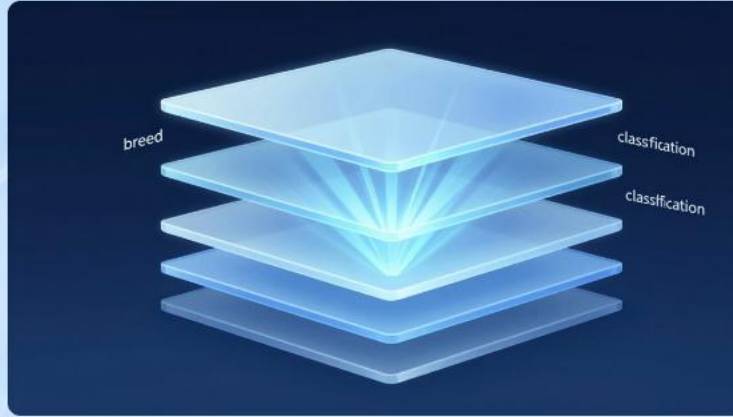
- Government breed conservation programs suffer from inaccurate data, hindering effective policy making.
- Accurate ID enables targeted conservation
 - Improves subsidy allocation efficiency
 - Supports national livestock database

Proposed Solution: BovID AI System



Mobile-Based AI Solution

- Identifies cattle and buffalo breeds instantly
- Uses face, body, horns, hump images
- Includes optional location metadata



Explainable AI Technology

- Provides confidence scores for predictions
- Explains why specific breed was identified
- Shows visual evidence and breed details



Designed for Rural India

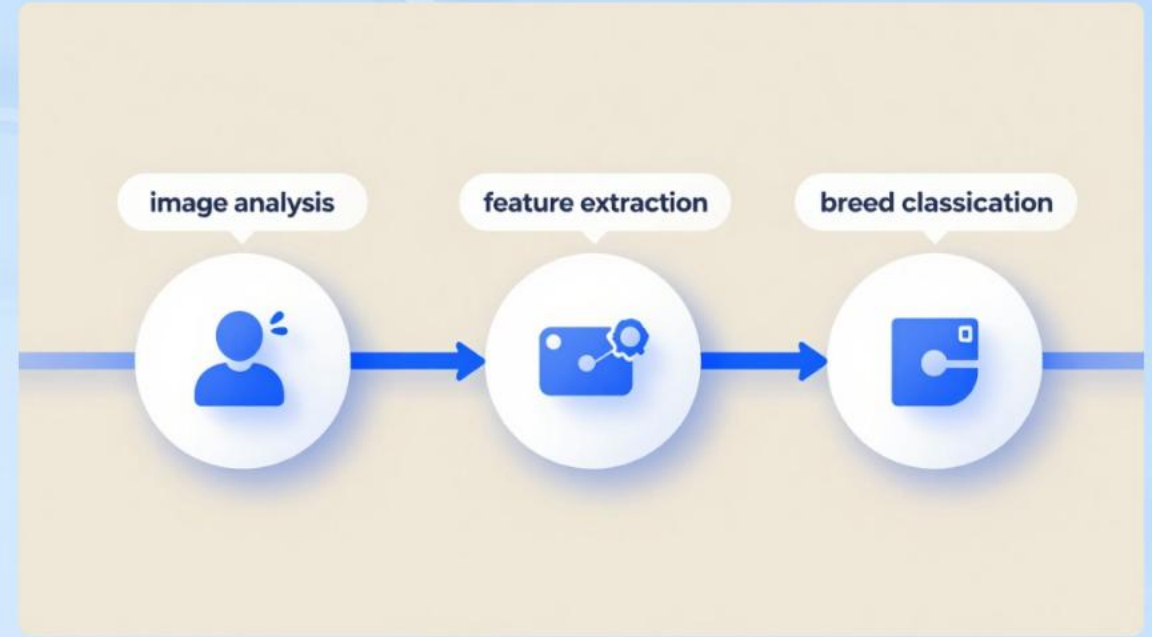
- Works on low-end smartphones
- Optimized for low-internet conditions
- Supports regional languages

How It Works



Capture Animal Images

- User captures or uploads animal photos
- Multiple angles: face, body, horns, hump
- App guides optimal image capture



AI Analysis & Prediction

- AI analyzes visual breed features
- Model predicts breed with confidence score
- Explainable AI shows reasoning process

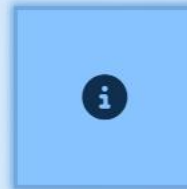
Key Features

Image-Based Detection

Advanced computer vision identifies breeds from face, body, horns, and hump characteristics with high accuracy.

Regional Language Support

Built for non-technical users with regional language interfaces and simple navigation.



Comprehensive Breed Info

Provides origin, milk yield, traits, and characteristics for each identified breed instantly.

Explainable AI Output

Shows visual evidence and reasoning why specific breed was identified, building user trust.

Innovation & Uniqueness



IN Indian Breed Specialization

- Trained specifically on 74+ Indian cattle & buffalo breeds
- Covers indigenous and recognized native breeds
- Preserves India's livestock heritage



Explainable AI Technology

- Not a black-box – shows reasoning process
- Builds farmer trust through transparency
- Provides visual evidence for predictions



Mobile-Optimized Model

- Lightweight model under 50MB
- Works offline for low-connectivity areas
- Optimized for low-end smartphones

Technology Stack (High-Level)

BovID combines cutting-edge mobile technology with optimized AI models, designed for scalability and performance in rural environments.

Component	Technology	Purpose	Key Feature
Frontend	Android App	User interface & capture	Offline-capable
AI Model	CNN Vision Model	Breed identification	Explainable predictions
Optimization	Mobile Inference	Fast on-device processing	< 3 sec response
Data Source	Govt + Field Images	Training dataset	74+ Indian breeds
Deployment	On-device + Cloud	Flexible connectivity	Works offline/online

Challenges & Our Approach

BovID addresses real-world challenges through innovative solutions, ensuring reliable performance in diverse rural conditions.

Challenge	Our Solution	Impact
Similar-looking breeds	Multi-view image analysis	Improved accuracy
Poor image quality	Pre-inference image checks	Better input validation
Limited training data	Augmentation + transfer learning	Enhanced model robustness
Low internet connectivity	Offline inference capability	Uninterrupted service
Rural user adoption	Simple UI + regional languages	Higher acceptance rate

Impact & Benefits

10M+
Indian Farmers

85%+
Target Accuracy

3s
Inference Time

For Farmers

Empowering farmers with accurate breed identification for better breeding decisions and improved income.

- Correct breed identification
- Better breeding & income
- Access to rightful subsidies

For Government

Enabling accurate subsidy verification and supporting indigenous breed conservation programs.

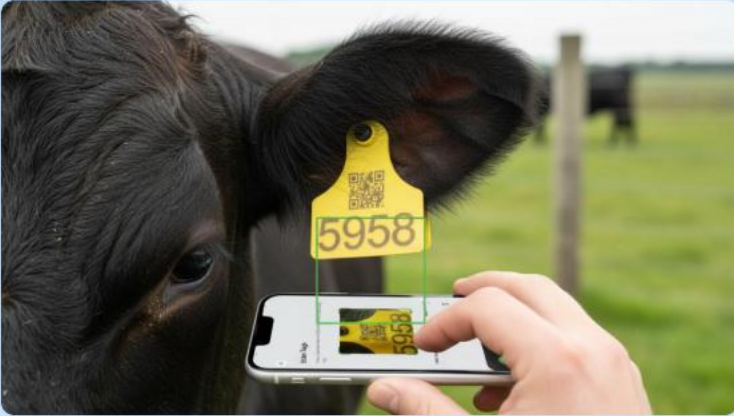
- Accurate subsidy verification
- Breed conservation support
- Policy implementation aid

For Researchers

Providing structured livestock data for scientific research and breed improvement programs.

- Structured livestock data
- Research insights
- Breed improvement support

Future Scope



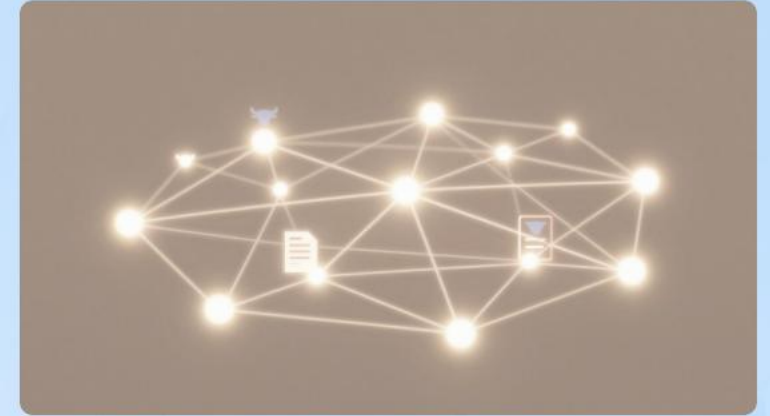
RFID / Ear-Tag Integration

- Digital livestock identification
- Automated breed tracking
- Health monitoring integration



Disease & Health Detection

- Early disease identification
- Health status monitoring
- Preventive care alerts



Blockchain Certification

- Tamper-proof breed certificates
- Digital verification system
- Trust and transparency

Success Metrics

85%

Identification Accuracy

3s

Inference Time

50MB

Model Size

Performance Targets

BovID aims to achieve industry-leading accuracy while maintaining fast response times and lightweight deployment.

- $\geq 85\%$ breed identification accuracy
- < 3 seconds inference time
- Model size < 50 MB

User Experience

Focus on positive field usability feedback and widespread adoption among rural farmers.

- Positive field usability feedback
- High user adoption rate
- Regional language satisfaction

Scalability Goals

Designed for national deployment with support for all recognized Indian cattle and buffalo breeds.

- Support for 74+ Indian breeds
- National deployment capability
- Government integration readiness

Final Pitch

Revolutionary Livestock Management

BovID empowers farmers with AI-driven, explainable breed identification — preserving India's livestock heritage while improving productivity.



Ready for Deployment

Field-tested technology ready for national rollout with government support.



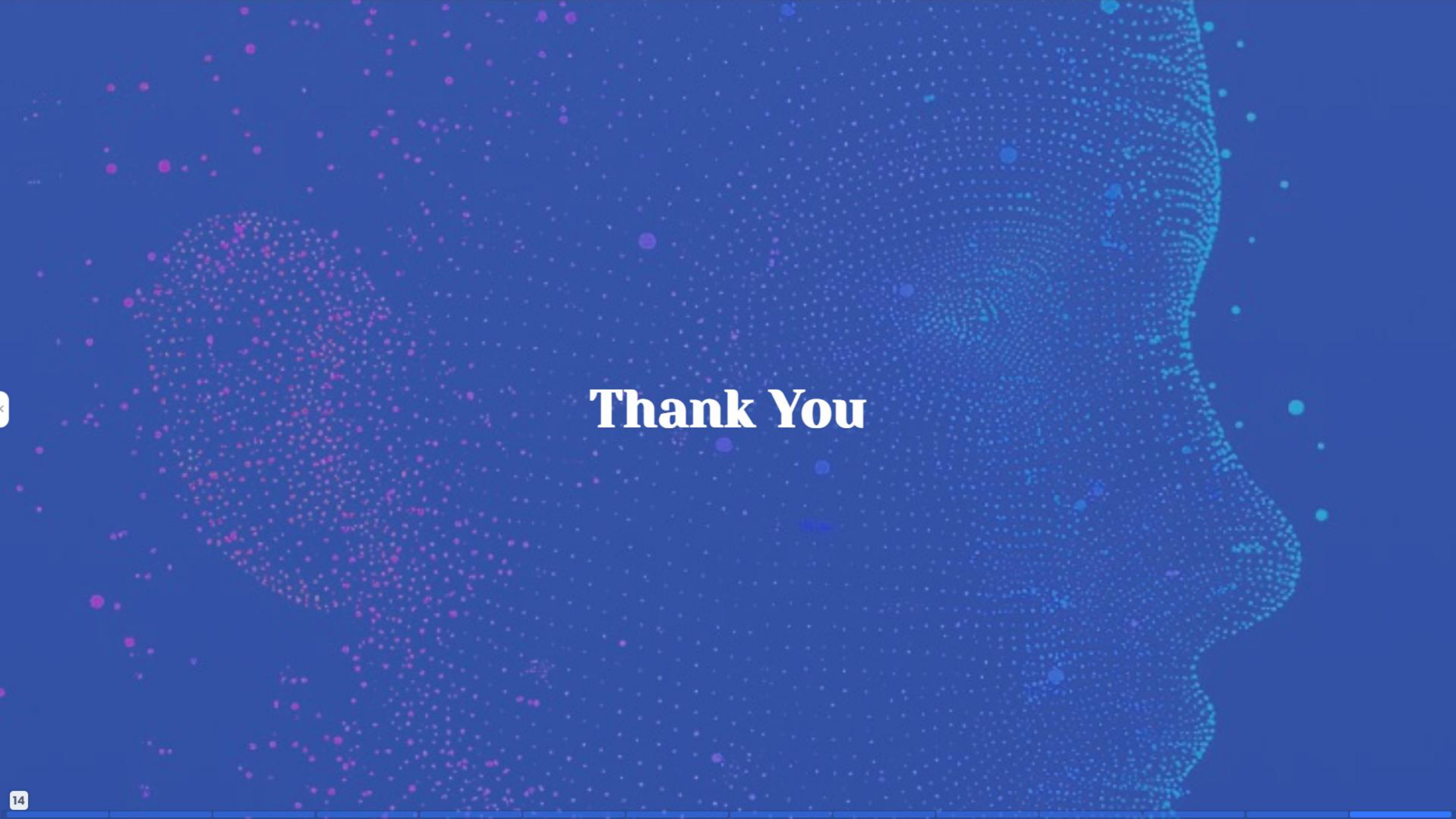
Partnership Opportunity

Seeking collaboration for nationwide implementation and breed conservation.



Sustainable Impact

Creating lasting positive change for farmers, government, and indigenous breeds.



Thank You